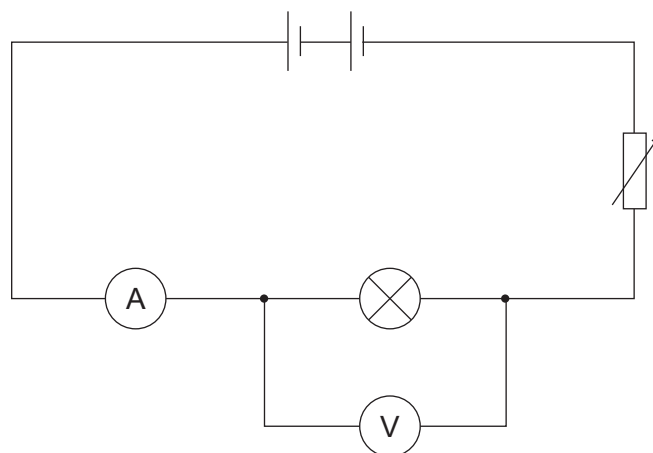


6. A student was investigating how current changes with voltage for various components. She used the following circuit to investigate a lamp.



She followed the method below.

1. Connect the circuit as shown in the diagram.
2. Set the variable resistor to give the lowest voltage and record the readings on the voltmeter and ammeter.
3. Alter the variable resistor to increase the voltage by 0.5 V.
4. Record the new readings on the voltmeter and ammeter.
5. Repeat steps 3 and 4.

(a) Some of the results are shown below.

Voltage (V)	Current (A)
0.0	0.00
2.5	0.65
3.5	0.90
4.5	1.16
5.0	1.20
5.5	1.26
6.5	1.36
8.0	1.46

(i) State the independent variable in this experiment.

[1]

.....

(ii) State the dependent variable in this experiment.

[1]

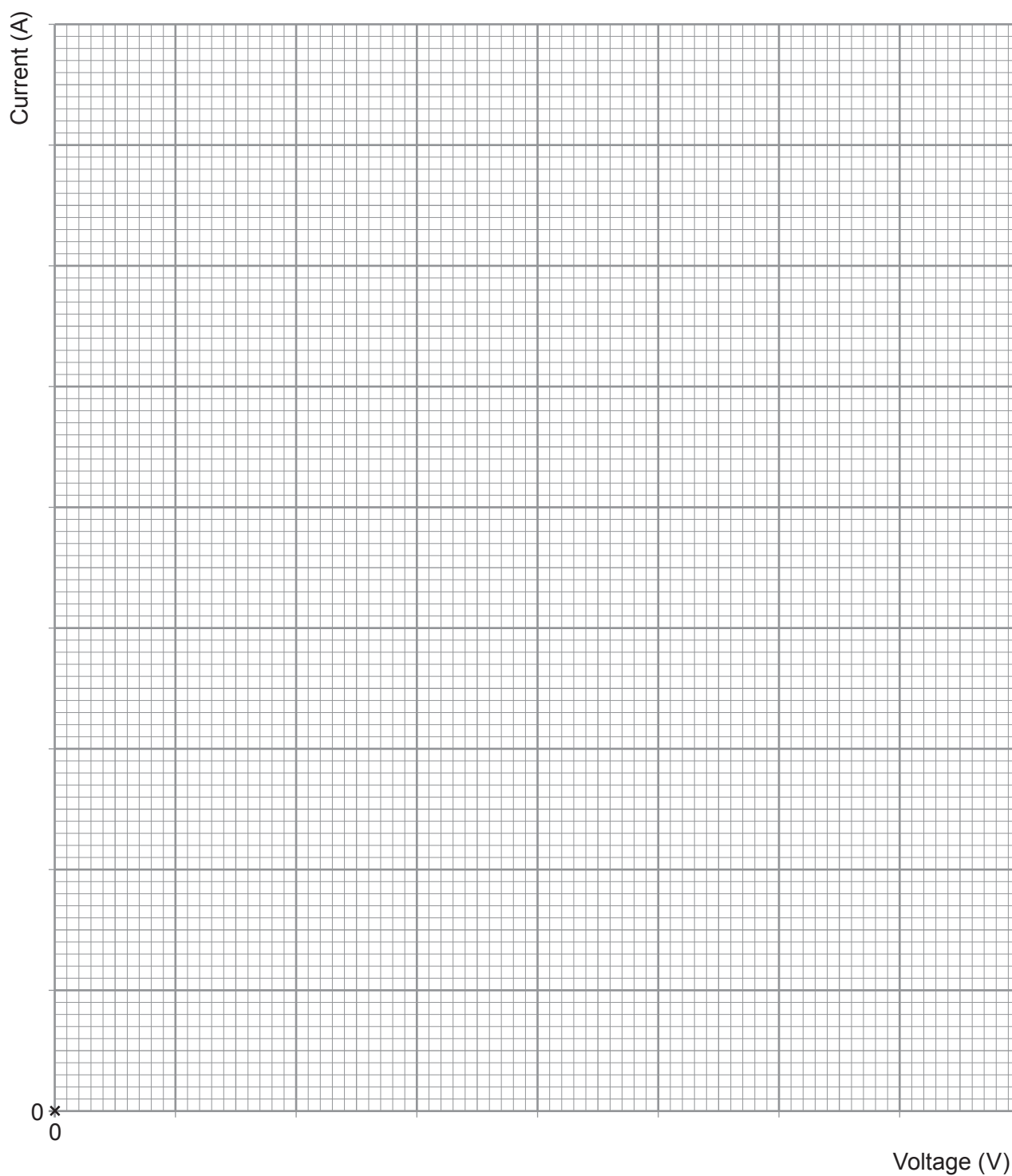
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(iii) Plot the results on the grid below and draw a suitable line.

[4]

Examiner
only



- (iv) Use your graph in (a)(iii) and the following equation

$$\text{voltage} = \text{current} \times \text{resistance}$$

to calculate the resistance of the lamp when the voltage is 7.2 V.

[3]

resistance = Ω

- (v) Explain whether the resistance of the lamp is constant throughout the experiment. [3]

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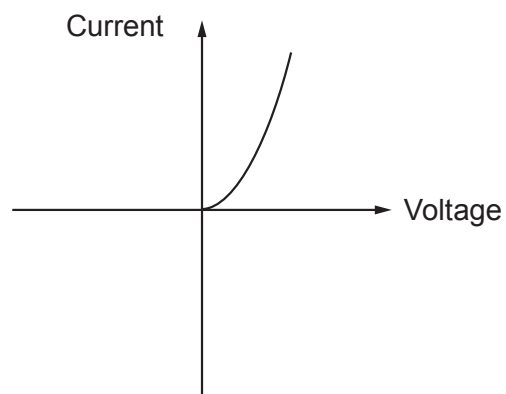
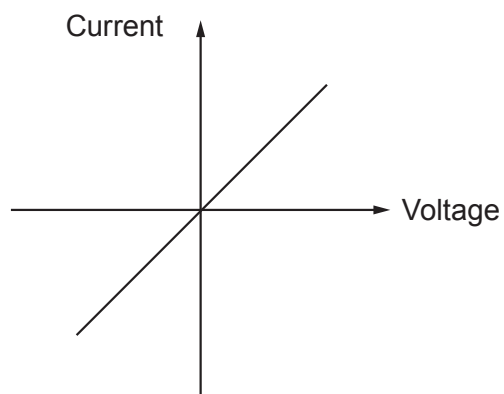
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- (b) The investigation was repeated with a resistor and then a diode.
The graphs of the results are shown below.



Use the graphs to compare the variation in resistance for these components.

[4]

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END OF PAPER





THE PERIODIC TABLE

Group

1 2

3 4 5 6 7 0

¹H Hydrogen 1										⁴He Helium 2									
</																			

Key

relative atomic mass

Ar

Symbol

Name

Z

atomic number